Current results for algebra2005

theta_m=0.8, theta_f=0.8, rec=50; remove training exfr0.00 edges

Train edges: mastery 48,160; forgetting 258,120; recommendation 21,500

N = 5, 10, ..., 50; Ep thresholds = 0.4, 0.5, 0.6, 0.7; planned seeds = 2024, 2025, 2026.

Cell = mean across available seed-level means +/- sample SD; a single result is shown without SD.

Bold = best; underline = second best among available models, ranked on unrounded means.

Cross-model rankings are provisional where seed coverage differs. CF baselines are deterministic.

Completed results included in this snapshot

Model	Seeds / evaluation	Section
2CKG4ER	2024,2025,2026	Ablation
No-mastery	2024,2025,2026	Ablation
No-forgetting	2024,2025,2026	Ablation
No-student-feature	2024,2025,2026	Ablation
No-exercise-feature	2024,2025,2026	Ablation
ID-only	2024,2025,2026	Ablation
EB-CF	deterministic	Comparison
SB-CF	deterministic	Comparison
CBF	deterministic	Comparison
TransE	2024,2025,2026	Comparison
TransE-adv	2024,2025,2026	Comparison
RotatE	-	Comparison
DistMult	-	Comparison
ComplEx	-	Comparison

Source: 4ER-New\Data_Fin_New_forgetting_edge_runs_rec50\drop_exfr0p00\algebra2005\forgetting_rec50_seed2024

Ablation | algebra2005

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	**0.790±0.001**	0.405±0.041	0.699±0.000	0.723±0.010	0.773±0.004	0.683±0.025
10	0.743±0.005	0.425±0.033	0.699±0.000	0.699±0.018	**0.745±0.003**	0.674±0.016
15	0.716±0.020	0.426±0.034	0.699±0.000	0.681±0.021	**0.721±0.012**	0.664±0.011
20	**0.712±0.017**	0.410±0.015	0.700±0.002	0.669±0.025	0.711±0.009	0.652±0.006
25	0.692±0.004	0.410±0.015	**0.703±0.003**	0.656±0.025	0.692±0.006	0.642±0.004
30	0.673±0.006	0.414±0.013	**0.708±0.003**	0.645±0.023	0.675±0.007	0.631±0.004
35	0.662±0.010	0.409±0.017	**0.711±0.003**	0.634±0.023	0.665±0.004	0.623±0.004
40	0.655±0.006	0.400±0.019	**0.715±0.001**	0.624±0.024	0.658±0.003	0.615±0.006
45	0.649±0.004	0.392±0.021	**0.715±0.001**	0.613±0.023	0.653±0.002	0.607±0.008
50	0.644±0.007	0.386±0.023	**0.714±0.002**	0.603±0.022	0.649±0.003	0.599±0.008

Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	**0.993±0.001**	0.975±0.002	0.991±0.000	0.990±0.001	0.991±0.002	0.986±0.006
10	0.988±0.001	0.975±0.001	**0.991±0.000**	0.989±0.003	0.986±0.001	0.985±0.004
15	0.985±0.001	0.975±0.001	**0.991±0.000**	0.988±0.003	0.984±0.001	0.985±0.004
20	0.984±0.000	0.974±0.001	**0.991±0.000**	0.987±0.004	0.984±0.001	0.985±0.004
25	0.985±0.000	0.974±0.001	**0.991±0.000**	0.987±0.003	0.984±0.000	0.986±0.003
30	0.985±0.001	0.974±0.001	**0.991±0.000**	0.987±0.003	0.984±0.000	0.985±0.003
35	0.985±0.001	0.974±0.001	**0.991±0.000**	0.987±0.004	0.984±0.000	0.986±0.003
40	0.985±0.001	0.973±0.002	**0.991±0.000**	0.987±0.003	0.985±0.000	0.986±0.003
45	0.985±0.000	0.973±0.001	**0.991±0.000**	0.987±0.003	0.985±0.000	0.986±0.002
50	0.985±0.000	0.973±0.001	**0.990±0.001**	0.987±0.002	0.985±0.000	0.986±0.002

Ablation | algebra2005

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Ep, response threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	-0.162±0.028	0.048±0.086	**0.176±0.005**	-0.044±0.018	-0.110±0.042	0.014±0.035
10	0.001±0.034	0.089±0.021	**0.228±0.006**	0.007±0.019	0.036±0.025	0.042±0.025
15	-0.008±0.154	0.120±0.031	**0.253±0.009**	0.055±0.008	0.059±0.049	0.091±0.020
20	0.042±0.041	0.092±0.018	**0.237±0.032**	0.088±0.005	0.039±0.034	0.087±0.002
25	-0.063±0.067	0.116±0.031	**0.169±0.094**	0.111±0.011	-0.025±0.030	0.092±0.006
30	0.007±0.048	0.080±0.019	0.027±0.134	0.112±0.011	0.075±0.103	**0.115±0.008**
35	**0.178±0.065**	0.093±0.012	0.025±0.042	0.123±0.008	0.092±0.068	0.117±0.032
40	**0.209±0.021**	0.039±0.041	-0.011±0.044	0.133±0.009	0.156±0.056	0.125±0.022
45	0.205±0.034	0.050±0.056	0.110±0.017	0.135±0.011	**0.211±0.005**	0.135±0.024
50	0.197±0.045	0.035±0.049	0.147±0.055	0.125±0.008	**0.214±0.012**	0.132±0.025

Ep, response threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	-0.169±0.025	-0.007±0.127	**0.172±0.005**	-0.061±0.020	-0.154±0.078	-0.005±0.043
10	-0.009±0.040	0.063±0.009	**0.227±0.006**	-0.038±0.032	0.005±0.021	0.015±0.031
15	-0.018±0.167	0.108±0.029	**0.252±0.009**	0.012±0.022	0.052±0.050	0.068±0.025
20	0.025±0.039	0.080±0.023	**0.237±0.032**	0.060±0.012	0.037±0.034	0.068±0.003
25	-0.066±0.064	0.102±0.040	**0.169±0.094**	0.088±0.013	-0.043±0.036	0.075±0.006
30	-0.006±0.065	0.065±0.026	0.026±0.133	0.092±0.013	0.063±0.099	**0.096±0.005**
35	**0.171±0.075**	0.075±0.017	0.020±0.048	0.103±0.010	0.088±0.069	0.099±0.033
40	**0.205±0.027**	0.019±0.040	-0.033±0.074	0.114±0.015	0.152±0.054	0.104±0.023
45	0.202±0.031	0.038±0.051	0.106±0.013	0.112±0.015	**0.209±0.006**	0.115±0.028
50	0.194±0.042	0.025±0.050	0.147±0.056	0.098±0.014	**0.212±0.013**	0.107±0.027

Ablation | algebra2005

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Ep, response threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	-0.299±0.010	-0.071±0.169	**0.112±0.002**	-0.147±0.030	-0.284±0.065	-0.075±0.052
10	-0.173±0.059	-0.011±0.010	**0.170±0.009**	-0.164±0.041	-0.191±0.025	-0.091±0.044
15	-0.150±0.238	0.042±0.051	**0.200±0.006**	-0.136±0.038	-0.086±0.093	-0.056±0.035
20	-0.087±0.132	0.001±0.029	**0.192±0.039**	-0.101±0.035	-0.100±0.108	-0.065±0.013
25	-0.284±0.030	0.048±0.064	**0.125±0.086**	-0.070±0.014	-0.238±0.039	-0.062±0.007
30	-0.260±0.065	**0.010±0.035**	-0.183±0.125	-0.061±0.037	-0.155±0.141	-0.032±0.018
35	-0.016±0.179	**0.018±0.030**	-0.305±0.095	-0.048±0.026	-0.095±0.025	-0.029±0.046
40	**0.102±0.053**	-0.033±0.042	-0.404±0.008	-0.034±0.030	0.015±0.060	-0.033±0.045
45	**0.126±0.017**	-0.013±0.064	-0.321±0.069	-0.024±0.030	0.120±0.008	-0.018±0.055
50	0.130±0.041	-0.029±0.075	-0.269±0.074	-0.038±0.022	**0.148±0.006**	-0.019±0.055

Ep, response threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-only
5	-0.363±0.013	-0.130±0.155	**-0.008±0.006**	-0.222±0.037	-0.366±0.052	-0.172±0.055
10	-0.304±0.055	-0.153±0.042	**-0.003±0.010**	-0.269±0.045	-0.289±0.025	-0.224±0.037
15	-0.313±0.143	-0.086±0.051	**0.002±0.006**	-0.256±0.031	-0.273±0.047	-0.220±0.040
20	-0.330±0.032	-0.146±0.021	**-0.056±0.080**	-0.240±0.029	-0.303±0.087	-0.235±0.029
25	-0.429±0.032	**-0.061±0.088**	-0.152±0.101	-0.230±0.009	-0.406±0.029	-0.246±0.016
30	-0.373±0.061	**-0.127±0.057**	-0.372±0.108	-0.234±0.021	-0.309±0.114	-0.230±0.017
35	-0.136±0.172	**-0.131±0.047**	-0.400±0.033	-0.230±0.021	-0.293±0.031	-0.234±0.042
40	**-0.044±0.127**	-0.195±0.048	-0.428±0.019	-0.228±0.017	-0.240±0.067	-0.234±0.042
45	**-0.016±0.042**	-0.181±0.062	-0.335±0.054	-0.223±0.023	-0.149±0.049	-0.230±0.045
50	**-0.039±0.130**	-0.194±0.074	-0.309±0.035	-0.229±0.009	-0.111±0.108	-0.229±0.044

Comparison | algebra2005

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Adaptability (Ada)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.222	0.450	0.274	0.730±0.005	0.725±0.005	-	-	-	**0.790±0.001**
10	0.220	0.515	0.269	0.701±0.005	0.711±0.008	-	-	-	**0.743±0.005**
15	0.219	0.498	0.269	0.672±0.005	0.689±0.004	-	-	-	**0.716±0.020**
20	0.217	0.468	0.271	0.653±0.006	0.676±0.006	-	-	-	**0.712±0.017**
25	0.218	0.441	0.272	0.637±0.008	0.668±0.008	-	-	-	**0.692±0.004**
30	0.217	0.418	0.272	0.624±0.009	0.659±0.008	-	-	-	**0.673±0.006**
35	0.217	0.398	0.273	0.613±0.008	0.654±0.009	-	-	-	**0.662±0.010**
40	0.216	0.381	0.275	0.604±0.009	0.649±0.010	-	-	-	**0.655±0.006**
45	0.216	0.366	0.275	0.594±0.009	0.642±0.010	-	-	-	**0.649±0.004**
50	0.216	0.355	0.274	0.586±0.009	0.633±0.010	-	-	-	**0.644±0.007**

Novelty (NOV)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.957	0.975	0.993	0.991±0.001	**0.996±0.001**	-	-	-	0.993±0.001
10	0.961	0.979	0.994	0.992±0.000	**0.994±0.000**	-	-	-	0.988±0.001
15	0.964	0.982	**0.995**	0.994±0.000	0.993±0.001	-	-	-	0.985±0.001
20	0.964	0.981	0.994	**0.995±0.000**	0.994±0.000	-	-	-	0.984±0.000
25	0.965	0.981	0.991	**0.994±0.000**	0.993±0.001	-	-	-	0.985±0.000
30	0.966	0.981	0.990	**0.993±0.000**	0.993±0.001	-	-	-	0.985±0.001
35	0.965	0.980	0.990	**0.993±0.000**	0.992±0.000	-	-	-	0.985±0.001
40	0.965	0.979	0.990	**0.993±0.000**	0.992±0.001	-	-	-	0.985±0.001
45	0.965	0.980	0.990	**0.993±0.000**	0.991±0.001	-	-	-	0.985±0.000
50	0.964	0.980	0.990	**0.993±0.000**	0.991±0.000	-	-	-	0.985±0.000

Comparison | algebra2005

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Ep, response threshold 0.4

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.037	0.047	**0.136**	-0.102±0.012	-0.054±0.018	-	-	-	-0.162±0.028
10	0.021	**0.163**	0.157	-0.008±0.004	0.009±0.015	-	-	-	0.001±0.034
15	0.002	**0.188**	0.164	-0.003±0.015	-0.018±0.023	-	-	-	-0.008±0.154
20	-0.019	**0.164**	0.154	0.026±0.023	0.058±0.027	-	-	-	0.042±0.041
25	-0.018	**0.171**	0.164	0.099±0.010	0.101±0.017	-	-	-	-0.063±0.067
30	-0.032	**0.167**	0.146	0.144±0.025	0.137±0.005	-	-	-	0.007±0.048
35	-0.021	0.150	0.155	**0.189±0.004**	0.145±0.012	-	-	-	0.178±0.065
40	-0.045	0.147	0.141	0.185±0.003	0.154±0.019	-	-	-	**0.209±0.021**
45	-0.015	0.146	0.139	0.169±0.011	0.163±0.014	-	-	-	**0.205±0.034**
50	-0.026	0.154	0.128	0.160±0.044	0.097±0.019	-	-	-	**0.197±0.045**

Ep, response threshold 0.5

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.039	0.041	**0.101**	-0.116±0.011	-0.061±0.017	-	-	-	-0.169±0.025
10	0.019	**0.159**	0.121	-0.041±0.015	-0.010±0.017	-	-	-	-0.009±0.040
15	-0.001	**0.183**	0.136	-0.045±0.022	-0.042±0.029	-	-	-	-0.018±0.167
20	-0.021	**0.156**	0.130	-0.014±0.016	0.026±0.030	-	-	-	0.025±0.039
25	-0.021	**0.162**	0.136	0.063±0.015	0.075±0.021	-	-	-	-0.066±0.064
30	-0.032	**0.161**	0.120	0.114±0.033	0.119±0.002	-	-	-	-0.006±0.065
35	-0.024	0.136	0.127	0.167±0.009	0.133±0.011	-	-	-	**0.171±0.075**
40	-0.045	0.137	0.109	0.165±0.004	0.140±0.017	-	-	-	**0.205±0.027**
45	-0.017	0.139	0.102	0.147±0.010	0.146±0.013	-	-	-	**0.202±0.031**
50	-0.030	0.146	0.092	0.134±0.044	0.078±0.021	-	-	-	**0.194±0.042**

Comparison | algebra2005

Mean +/- SD across available seeds; one result has no SD. Missing models are marked '-'.

Ep, response threshold 0.6

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.042	0.002	**0.045**	-0.209±0.008	-0.156±0.017	-	-	-	-0.299±0.010
10	0.011	**0.082**	0.049	-0.198±0.009	-0.134±0.011	-	-	-	-0.173±0.059
15	-0.011	**0.135**	0.061	-0.248±0.020	-0.217±0.025	-	-	-	-0.150±0.238
20	-0.034	**0.110**	0.067	-0.229±0.015	-0.187±0.038	-	-	-	-0.087±0.132
25	-0.032	**0.128**	0.066	-0.167±0.023	-0.129±0.012	-	-	-	-0.284±0.030
30	-0.043	**0.118**	0.046	-0.098±0.041	-0.024±0.019	-	-	-	-0.260±0.065
35	-0.034	**0.095**	0.059	-0.024±0.015	0.022±0.025	-	-	-	-0.016±0.179
40	-0.060	**0.102**	0.049	0.043±0.010	0.037±0.041	-	-	-	0.102±0.053
45	-0.031	0.108	0.033	0.035±0.010	0.052±0.032	-	-	-	**0.126±0.017**
50	-0.046	0.115	0.022	0.035±0.056	-0.019±0.039	-	-	-	**0.130±0.041**

Ep, response threshold 0.7

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.078	-0.101	**-0.014**	-0.286±0.008	-0.241±0.011	-	-	-	-0.363±0.013
10	-0.036	-0.066	**-0.033**	-0.268±0.014	-0.244±0.020	-	-	-	-0.304±0.055
15	-0.075	**-0.022**	-0.045	-0.316±0.019	-0.310±0.020	-	-	-	-0.313±0.143
20	-0.106	-0.046	**-0.038**	-0.293±0.016	-0.279±0.018	-	-	-	-0.330±0.032
25	-0.099	**-0.023**	-0.074	-0.253±0.023	-0.250±0.013	-	-	-	-0.429±0.032
30	-0.120	**-0.013**	-0.099	-0.185±0.041	-0.171±0.028	-	-	-	-0.373±0.061
35	-0.135	**-0.019**	-0.096	-0.127±0.020	-0.174±0.031	-	-	-	-0.136±0.172
40	-0.157	**-0.002**	-0.116	-0.079±0.019	-0.168±0.043	-	-	-	-0.044±0.127
45	-0.136	**0.010**	-0.125	-0.102±0.002	-0.145±0.039	-	-	-	-0.016±0.042
50	-0.167	**0.029**	-0.137	-0.123±0.061	-0.215±0.037	-	-	-	-0.039±0.130